

Disproof of Conjecture 00000007986: The Covering-Distance Square-Root Law for Codes with 2-Transitive Automorphism Group

TLMC submission — branch `submission/00000007986`

September 2026

Abstract

The main clause of Conjecture 00000007986 asserts the inequality $\rho(C) \leq n - \sqrt{nd}$ for every code C whose automorphism group is nontrivial and 2-transitive. We disprove this by explicit counterexample: the binary repetition codes $C_4 = \{0000, 1111\} \subset \mathbb{F}_2^4$ and $C_2 = \{00, 11\} \subset \mathbb{F}_2^2$. Both satisfy the hypothesis (their automorphism groups contain S_4 resp. S_2 acting by coordinate permutations, which is nontrivial and 2-transitive), yet $\rho(C_4) = 2 > 0 = 4 - \sqrt{16}$ and $\rho(C_2) = 1 > 0 = 2 - \sqrt{4}$. All numbers are verified by exhaustive enumeration (`reproduce.py`) and by a zero-axiom Lean 4 formalization (`lean4/`).

1 Scope of the disproof

Conjecture 00000007986 contains a main inequality together with ancillary clauses (equality for Reed–Solomon type affine group codes, an orbit–rank correspondence, and a transitivity criterion for design strength). **We refute only the literal statement of the main inequality**

$$\rho(C) \leq n - \sqrt{nd},$$

under its own hypothesis that C has a nontrivial 2-transitive automorphism group. The ancillary clauses are not the target of this disproof.

2 Definitions

Definition 1 (Binary code). A *code* of length n is a subset $C \subseteq \mathbb{F}_2^n$. For $x, y \in \mathbb{F}_2^n$ the *Hamming distance* is $d(x, y) = |\{i \in \{1, \dots, n\} : x_i \neq y_i\}|$.

Definition 2 (Minimum distance and covering radius). The *minimum distance* of a code with $|C| \geq 2$ is $d = \min\{d(c_1, c_2) : c_1 \neq c_2 \in C\}$. The *covering radius* is

$$\rho(C) = \max_{x \in \mathbb{F}_2^n} \min_{c \in C} d(x, c).$$

Definition 3 (Automorphism group, 2-transitivity). A permutation $\sigma \in S_n$ *acts* on $\mathbb{F}_2^n$ by $(\sigma \cdot x)_i = x_{\sigma(i)}$; this action preserves Hamming distance. The *automorphism group* of C is $\text{Aut}(C) = \{\sigma \in S_n : \sigma \cdot C = C\}$, where $\sigma \cdot C = \{\sigma \cdot c : c \in C\}$. The group $\text{Aut}(C)$ acts on the index set $\{1, \dots, n\}$ via the natural action of S_n ; it is *2-transitive* if this action is 2-transitive, and it is *nontrivial* if $\text{Aut}(C) \neq \{\text{id}\}$ (in particular $\text{Aut}(C) \supseteq A_n$ or S_n for our examples, which are 2-transitive for $n \geq 2$).

3 The counterexamples

Counterexample 1 (Primary: the repetition code C_4). Let $C_4 = \{0000, 1111\} \subset \mathbb{F}_2^4$.

- $n = 4$ and $d = 4$ (the two distinct codewords differ in all four coordinates).
- Hypothesis holds. Every $\sigma \in S_4$ fixes the all-zero word and maps 1111 to 1111, hence $\text{Aut}(C_4) = S_4$ (all 24 coordinate permutations stabilize the code set). S_4 is nontrivial and 2-transitive on $\{1, 2, 3, 4\}$.
- Covering radius. Enumerating all 16 vectors of $\mathbb{F}_2^4$ (Table 1) gives $\rho(C_4) = \max_x \min(d(x, 0000), d(x, 1111)) = 2$: a vector x of weight w has $d(x, 0000) = w$ and $d(x, 1111) = 4 - w$, so $\min = \min(w, 4 - w) \leq 2$, with equality at $w = 2$ (e.g. $x = 0011$).
- Violation. The conjectured bound is $4 - \sqrt{4 \cdot 4} = 4 - 4 = 0$, and the claim $\rho(C_4) \leq 0$ reads $2 \leq 0$, which is false.

Table 1: Exhaustive enumeration of $\mathbb{F}_2^4$ for $C_4 = \{0000, 1111\}$.

weight w	vectors x (representatives of this class)	$d(x, 0000)$	$d(x, 1111)$	min
0	0000	0	4	0
1	0001, 0010, 0100, 1000	1	3	1
2	0011, 0101, 0110, 1001, 1010, 1100	2	2	2
3	0111, 1011, 1101, 1110	3	1	1
4	1111	4	0	0

$$\rho(C_4) = \max = 2. \quad \text{RHS} = 4 - \sqrt{4 \cdot 4} = 0. \quad 2 \leq 0 \text{ is false.}$$

Counterexample 2 (Secondary: the repetition code C_2). Let $C_2 = \{00, 11\} \subset \mathbb{F}_2^2$.

- $n = 2, d = 2$; $\text{Aut}(C_2) = S_2$, which is nontrivial and 2-transitive, so the hypothesis holds.
- Enumerating all 4 vectors of $\mathbb{F}_2^2$: x of weight w has $\min(d(x, 00), d(x, 11)) = \min(w, 2 - w) \leq 1$, with equality at $w = 1$ (e.g. $x = 01$). Hence $\rho(C_2) = 1$.
- The conjectured bound is $2 - \sqrt{2 \cdot 2} = 0$, and the claim $1 \leq 0$ is false.

Table 2: Exhaustive enumeration of $\mathbb{F}_2^2$ for $C_2 = \{00, 11\}$.

weight w	vectors x	$d(x, 00)$	$d(x, 11)$	min
0	00	0	2	0
1	01, 10	1	1	1
2	11	2	0	0

$$\rho(C_2) = \max = 1. \quad \text{RHS} = 2 - \sqrt{2 \cdot 2} = 0. \quad 1 \leq 0 \text{ is false.}$$

4 Conclusion

Theorem 1 (Disproof of Conjecture 00000007986, main clause). *The inequality $\rho(C) \leq n - \sqrt{nd}$ does not hold for all codes with nontrivial 2-transitive automorphism group. Both $C_4 = \{0000, 1111\} \subset \mathbb{F}_2^4$ and $C_2 = \{00, 11\} \subset \mathbb{F}_2^2$ satisfy the hypothesis, yet $\rho(C_4) = 2 > 0 = 4 - \sqrt{16}$ and $\rho(C_2) = 1 > 0 = 2 - \sqrt{4}$.*

Proof. For a binary vector x of weight w and the repetition code $R_n = \{0^n, 1^n\}$ of length n , we have $d(x, 0^n) = w$ and $d(x, 1^n) = n - w$, hence $\min(w, n - w) \leq \lfloor n/2 \rfloor$ with equality for $w = \lfloor n/2 \rfloor$ when n is even. Thus $\rho(R_n) = n/2$ for even n , while the conjectured bound is $n - \sqrt{n \cdot n} = 0$. For $n = 4$: $\rho = 2 > 0$; for $n = 2$: $\rho = 1 > 0$. In both cases $\text{Aut}(R_n) = S_n$ is nontrivial and 2-transitive, so the hypothesis of the conjecture is satisfied and the inequality fails. $\square$

Remark on the gap. In both counterexamples the right-hand side is 0 while the true covering radius is $n/2 > 0$, so no rounding convention or alternate reading of the square root rescues the inequality; the failure is unambiguous.

5 Machine verification

- `reproduce.py` exhaustively enumerates all 16 vectors of $\mathbb{F}_2^4$ and all 4 vectors of $\mathbb{F}_2^2$, recomputes $\rho(C_4) = 2$, $\rho(C_2) = 1$ and the right-hand sides $4 - \sqrt{16} = 0$, $2 - \sqrt{4} = 0$ (exact integer arithmetic, since nd is a perfect square), verifies the automorphism-group claim (all 24 permutations of S_4 resp. both of S_2 stabilize the code), and asserts the violations.
- `lean4/` contains a Lean 4 (v4.33.1, core only) project whose theorems `rho_C4 : coverRad4 = 2` and `rho_C2 : coverRad2 = 1` evaluate the full 16-point (resp. 4-point) enumeration directly in the kernel via `rf1`, while the numeric refutations $\neg((2:\text{Int}) \leq (4:\text{Int}) - 4)$ (`refute_bound_4`) and $\neg((1:\text{Int}) \leq (2:\text{Int}) - 2)$ (`refute_bound_2`) — the bound values $4 - 4 = 0$ resp. $2 - 2 = 0$ are exact integers, since nd is a perfect square — are proved by `decide`. `Check.lean` prints `#print axioms` for each theorem: all four are axiom-free and no `sorry` is used.